

Initiating Search

November 10, 2025, 9:40 PM

• Search:

CAS Lexicon Search:

Gewald reaction - Preferred Concept

Search Tasks

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Exported: Retrieved Related Reaction Results + Filters (574)	 Reactions	View Results
Filtered By:		
Yield: 90-100%, 80-89%, 70-79%		

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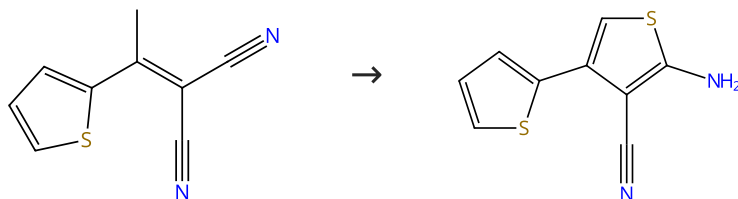
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Reactions (50)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (49)

 Suppliers (15)

31-025-CAS-18171340

Steps: 1 Yield: 100%

New conditions for the synthesis of thiophenes via the Knoevenagel/Gewald reaction sequence. Application to the synthesis of a multitargeted kinase inhibitor

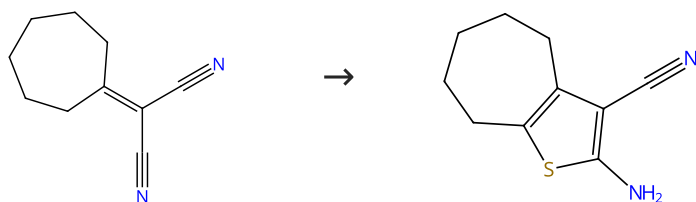
By: Barnes, David M.; et al

Tetrahedron (2006), 62(49), 11311-11319.

- 1.1 Reagents: Sulfur
Solvents: Tetrahydrofuran; rt → 35 °C
- 1.2 Reagents: Sodium bicarbonate
Solvents: Water; 1 h, 35 °C; 30 min, 35 °C

Scheme 2 (1 Reaction)

Steps: 1 Yield: 99%


 Suppliers (3)

 Suppliers (64)

31-025-CAS-18114238

Steps: 1 Yield: 99%

Activation of elemental sulfur by electrogenerated cyanomethyl anion: synthesis of substituted 2-aminothiophenes by the Gewald reaction

By: Feroci, Marta; et al

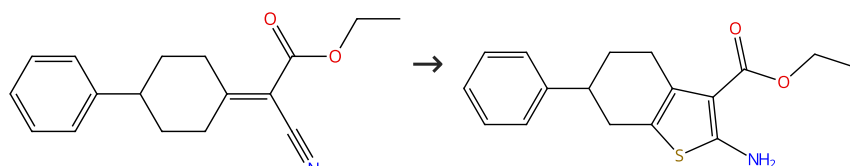
Advanced Synthesis & Catalysis (2008), 350(17), 2740-2746.

- 1.1 Reagents: S₈
Catalysts: Acetonitrile, Tetraethylammonium hexafluorophosphate
Solvents: Dimethylformamide; 1 min, rt; 1 h, rt

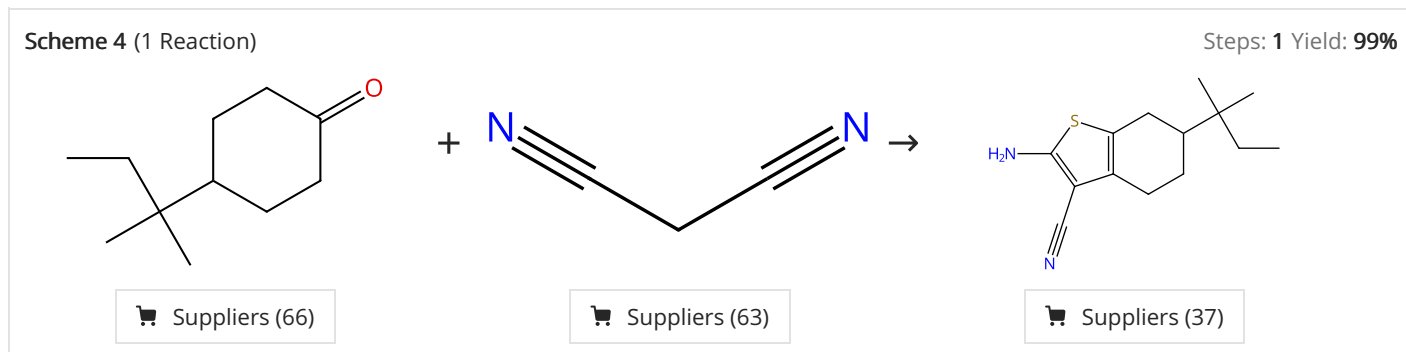
Experimental Protocols

Scheme 3 (1 Reaction)

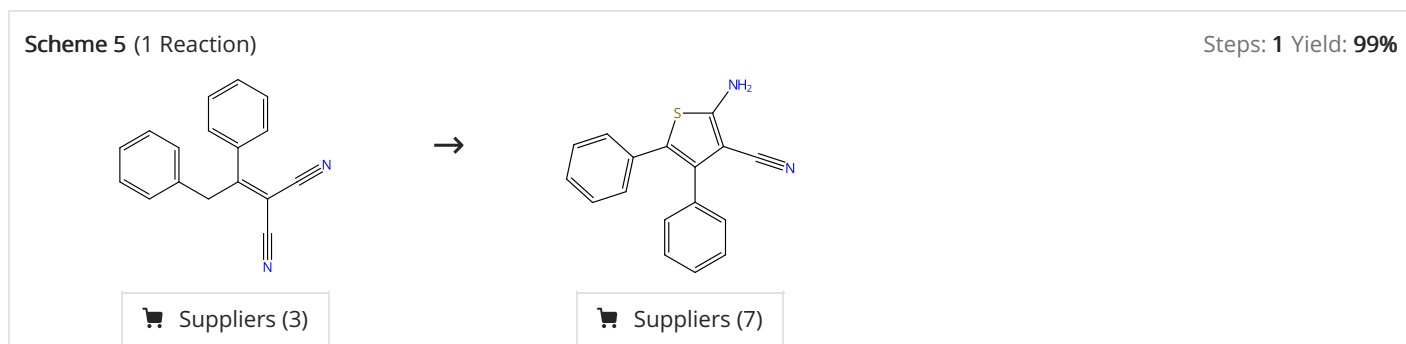
Steps: 1 Yield: 99%


 Suppliers (43)

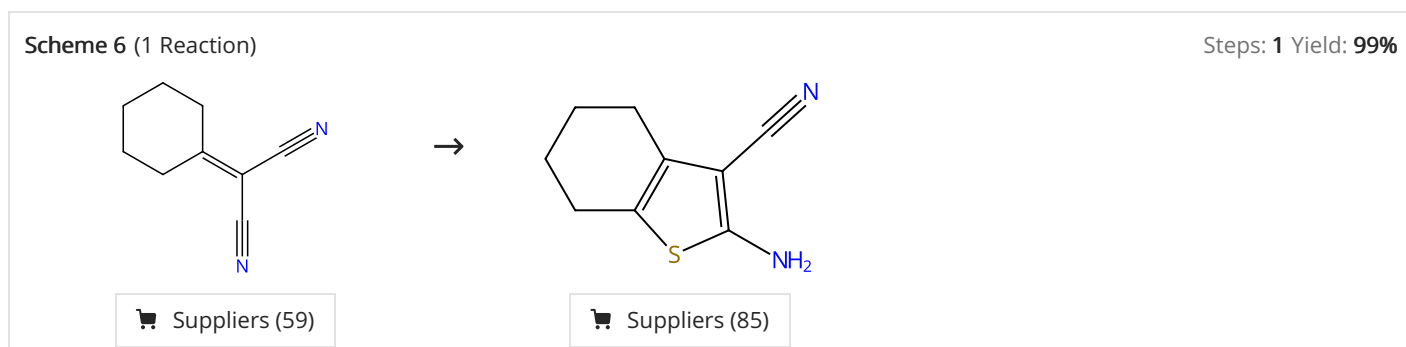
31-025-CAS-18163467	Steps: 1 Yield: 99%	Activation of elemental sulfur by electrogenerated cyanomethyl anion: synthesis of substituted 2-aminothiophenes by the Gewald reaction By: Feroci, Marta; et al Advanced Synthesis & Catalysis (2008), 350(17), 2740-2746.
1.1 Reagents: S ₈ Catalysts: Acetonitrile, Tetraethylammonium hexafluoro phosphate Solvents: Dimethylformamide; 1 min, rt; 120 min, 50 °C		
Experimental Protocols		



31-572-CAS-13427728	Steps: 1 Yield: 99%	Direct observation (NMR) of the efficacy of glucagon receptor antagonists in murine liver expressing the human glucagon receptor By: Cohen, Sheila M.; et al Bioorganic & Medicinal Chemistry (2006), 14(5), 1506-1517.
1.1 Reagents: Morpholine, Sulfur Solvents: Ethanol; 0 °C; 16 h, rt		



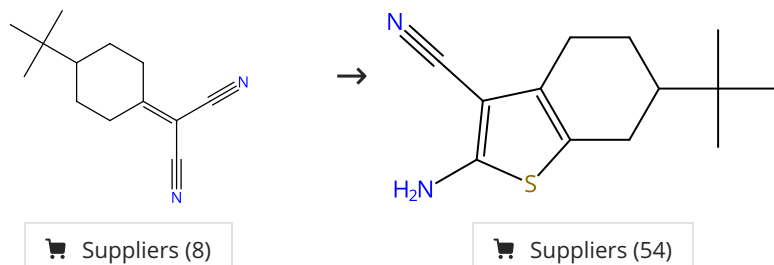
31-025-CAS-18261654	Steps: 1 Yield: 99%	New conditions for the synthesis of thiophenes via the Knoevenagel/Gewald reaction sequence. Application to the synthesis of a multitargeted kinase inhibitor By: Barnes, David M.; et al Tetrahedron (2006), 62(49), 11311-11319.
1.1 Reagents: Sulfur Solvents: Tetrahydrofuran; rt → 35 °C		
1.2 Reagents: Sodium bicarbonate Solvents: Water; 1 h, 35 °C; 30 min, 35 °C		



31-025-CAS-18119192	Steps: 1 Yield: 99%	Activation of elemental sulfur by electrogenerated cyanomethyl anion: synthesis of substituted 2-aminothiophenes by the Gewald reaction By: Feroci, Marta; et al Advanced Synthesis & Catalysis (2008), 350(17), 2740-2746.
1.1 Reagents: S ₈ Catalysts: Acetonitrile, Tetraethylammonium hexafluoro phosphate Solvents: Dimethylformamide; 1 min, rt; 1 h, rt		
Experimental Protocols		

Scheme 7 (1 Reaction)

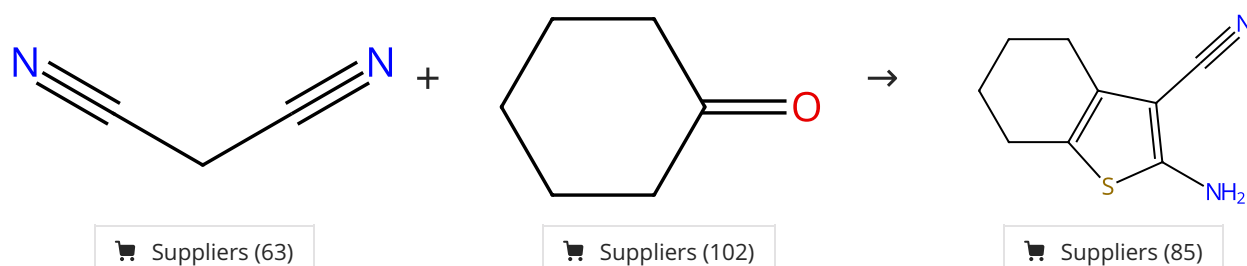
Steps: 1 Yield: 99%



31-025-CAS-18118430	Steps: 1 Yield: 99%	Activation of elemental sulfur by electrogenerated cyanomethyl anion: synthesis of substituted 2-aminothiophenes by the Gewald reaction By: Feroci, Marta; et al Advanced Synthesis & Catalysis (2008), 350(17), 2740-2746.
1.1 Reagents: S ₈ Catalysts: Acetonitrile, Tetraethylammonium hexafluoro phosphate Solvents: Dimethylformamide; 1 min, rt; 1 h, rt		
Experimental Protocols		

Scheme 8 (6 Reactions)

Steps: 1 Yield: 96-99%

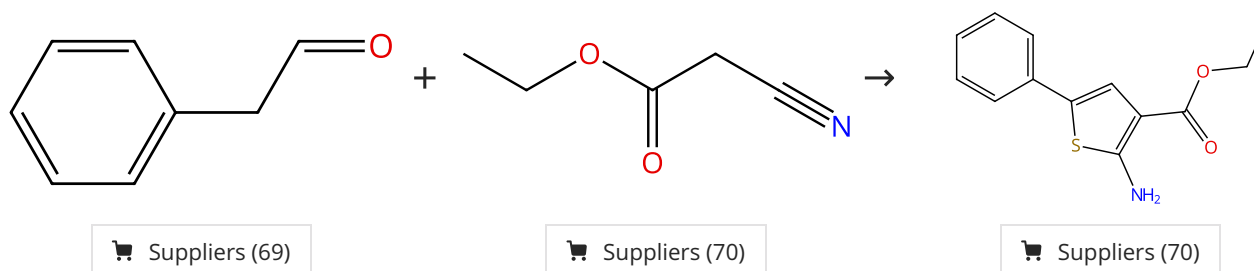


31-572-CAS-11693418	Steps: 1 Yield: 99%	Bovine serum albumin-catalyzed one-pot synthesis of 2-aminothiophenes via Gewald reaction By: Zhao, Dan-Dan; et al Journal of Molecular Catalysis B: Enzymatic (2013), 95, 29-35.
1.1 Reagents: Sulfur Solvents: Dimethylformamide; 4 h, 50 °C		
31-572-CAS-18750912	Steps: 1 Yield: 98%	A facile and environmentally benign polyethylene glycol 600-mediated method for the synthesis of densely functionalized 2-aminothiophene derivatives under ultrasonication By: Akbarzadeh, Ali; et al Green Chemistry Letters and Reviews (2017), 10(4), 315-323.
1.1 Reagents: S ₈ Solvents: Polyethylene glycol; 10 min, rt		
Experimental Protocols		
31-614-CAS-24573922	Steps: 1 Yield: 97%	Piperazine-Grafted Magnetic Reduced Graphene Oxide (Fe₃O₄@rGO-NH) as a Reusable Heterogeneous Catalyst for Gewald Three-Component Reaction By: Rezaei-Seresht, Esmail; et al Polycyclic Aromatic Compounds (2021), 41(9), 1944-1952.
1.1 Reagents: S ₈ Catalysts: Piperazine (reaction product with reduced graphene oxide) Solvents: Ethanol; 6 h, 85 °C		
Experimental Protocols		

31-572-CAS-12632838 Steps: 1 Yield: 97% 1.1 Reagents: S₈ Catalysts: L-Proline Solvents: Dimethylformamide; 10 h, 60 °C Experimental Protocols	An efficient one-pot synthesis of substituted 2-aminothiophenes via three-component Gewald reaction catalyzed by L-proline By: Wang, Tao; et al Synlett (2010), (9), 1351-1354.
31-614-CAS-38727532 Steps: 1 Yield: 96% 1.1 Reagents: S₈ Catalysts: Boric acid, compd. with piperidine (1:1) Solvents: Ethanol, Water; 25 min, 100 °C Experimental Protocols	Truly Catalytic Gewald Synthesis of 2- Aminothiophenes Using Piperidinium Borate (Pip Borate), a Conjugate Acid-Base Pair By: Gavali, Kanchan D.; et al SynOpen (2023), 7(4), 674-679.
31-572-CAS-13343857 Steps: 1 Yield: 96% 1.1 Reagents: S₈ Solvents: Diethylamine, Water; 5 min, 25 °C Experimental Protocols	Convenient synthesis of 2-aminothiophene derivatives by acceleration of Gewald reaction under ultrasonic aqueous conditions By: Mojtahedi, Mohammad M.; et al Synthetic Communications (2010), 40(14), 2067-2074.

Scheme 9 (1 Reaction)

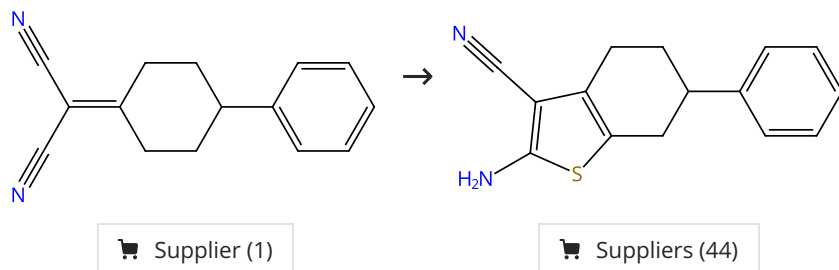
Steps: 1 Yield: 99%



31-572-CAS-11917756 Steps: 1 Yield: 99% 1.1 Reagents: Morpholine, S₈ Solvents: Ethanol; 20 min, 70 °C Experimental Protocols	Microwave-assisted synthesis of 5-substituted 2-aminothiophenes starting from arylacetaldehydes By: Revelant, Germain; et al Synthesis (2011), (18), 2935-2940.
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Scheme 10 (1 Reaction)

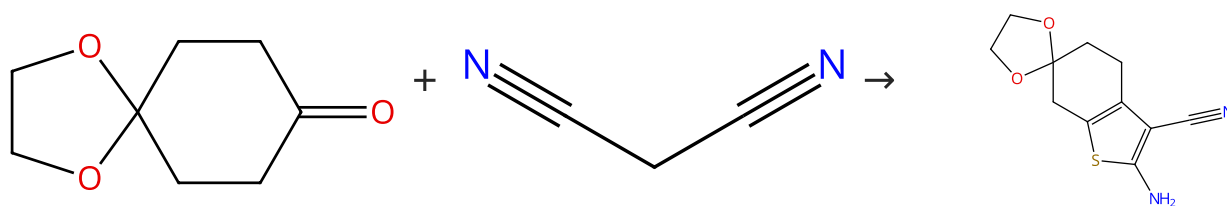
Steps: 1 Yield: 98%



31-025-CAS-18237410 Steps: 1 Yield: 98% 1.1 Reagents: S₈ Catalysts: Acetonitrile, Tetraethylammonium hexafluoro phosphate Solvents: Dimethylformamide; 1 min, rt; 1 h, rt Experimental Protocols	Activation of elemental sulfur by electrogenerated cyanomethyl anion: synthesis of substituted 2-aminothiophenes by the Gewald reaction By: Feroci, Marta; et al Advanced Synthesis & Catalysis (2008), 350(17), 2740-2746.
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Scheme 11 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (98)

Suppliers (63)

Suppliers (4)

31-572-CAS-6342042

Steps: 1 Yield: 98%

An efficient one-pot synthesis of substituted 2-aminothiophenes via three-component Gewald reaction catalyzed by L-proline

By: Wang, Tao; et al

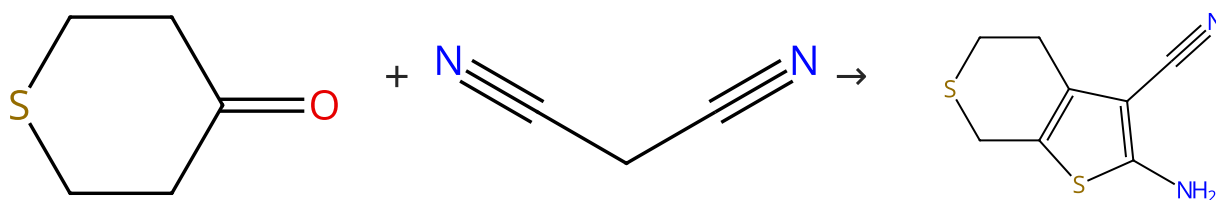
Synlett (2010), (9), 1351-1354.

1.1 Reagents: S₈
Catalysts: L-Proline
Solvents: Dimethylformamide; 10 h, 60 °C

Experimental Protocols

Scheme 12 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (97)

Suppliers (63)

Suppliers (24)

31-572-CAS-4471528

Steps: 1 Yield: 98%

Efficient three-component Gewald reactions under Et₃N/H₂O conditions

By: Abaee, M. Saeed; et al

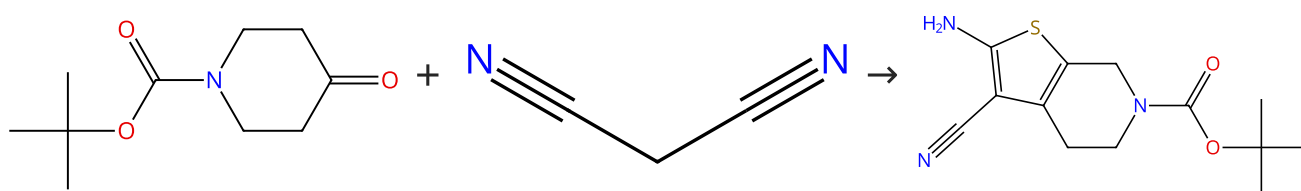
Journal of Sulfur Chemistry (2014), 35(3), 261-269.

1.1 Reagents: Triethylamine, Sulfur
Solvents: Water; 4 h, rt

Experimental Protocols

Scheme 13 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (55)

Suppliers (63)

Suppliers (59)

31-572-CAS-9776196

Steps: 1 Yield: 98%

An efficient one-pot synthesis of substituted 2-aminothiophenes via three-component Gewald reaction catalyzed by L-proline

By: Wang, Tao; et al

Synlett (2010), (9), 1351-1354.

1.1 Reagents: S₈
Catalysts: L-Proline
Solvents: Dimethylformamide; 10 h, 60 °C

Experimental Protocols

Scheme 14 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (3)

Suppliers (5)

31-025-CAS-18219519

Steps: 1 Yield: 98%

New conditions for the synthesis of thiophenes via the Knoevenagel/Gewald reaction sequence. Application to the synthesis of a multitargeted kinase inhibitor

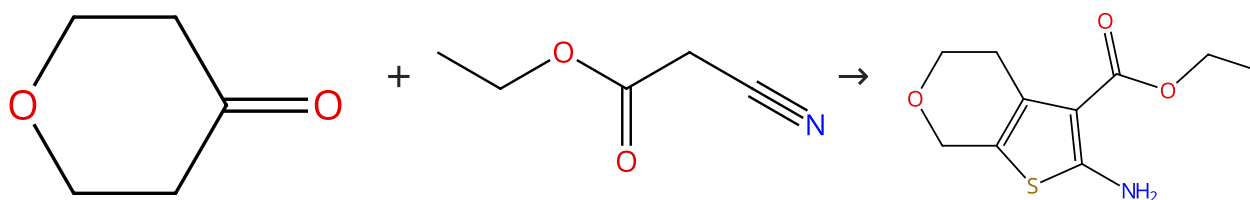
By: Barnes, David M.; et al

Tetrahedron (2006), 62(49), 11311-11319.

- 1.1 Reagents: Sulfur
Solvents: Tetrahydrofuran; rt → 35 °C
- 1.2 Reagents: Sodium bicarbonate
Solvents: Water; 1 h, 35 °C; 30 min, 35 °C

Scheme 15 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (93)

Suppliers (70)

Suppliers (59)

31-572-CAS-5894327

Steps: 1 Yield: 98%

Efficient three-component Gewald reactions under Et₃N/H₂O conditions

By: Abaee, M. Saeed; et al

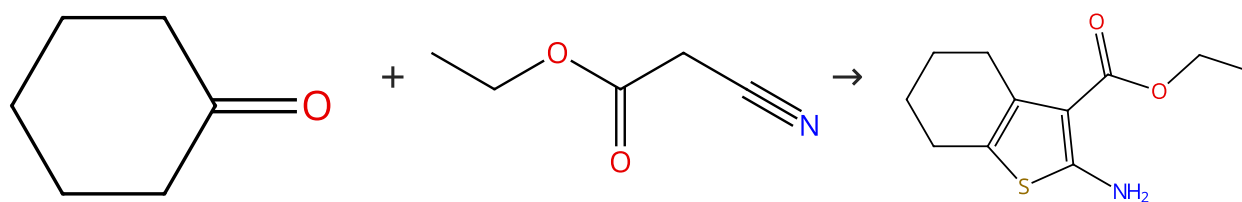
Journal of Sulfur Chemistry (2014), 35(3), 261-269.

- 1.1 Reagents: Triethylamine, Sulfur
Solvents: Water; 5 h, rt

Experimental Protocols

Scheme 16 (4 Reactions)

Steps: 1 Yield: 95-98%



Suppliers (102)

Suppliers (70)

Suppliers (77)

31-572-CAS-16138435

Steps: 1 Yield: 98%

Synthesis and antitumor activity evaluation of novel tetrahydro-[4,5]benzothieno[2,3-d]pyrimidinone derivatives

By: Chen, Yu; et al

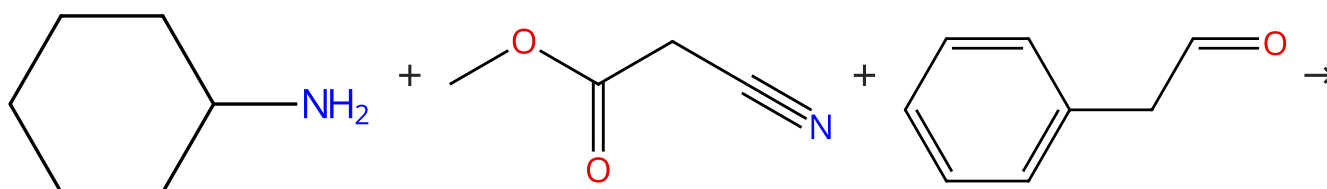
Youji Huaxue (2014), 34(11), 2362-2369.

- 1.1 Reagents: Morphine, S₈
Solvents: Ethanol; rt; 12 h, 60 °C

31-572-CAS-6957247 Steps: 1 Yield: 97% 1.1 Reagents: S₈ Solvents: Diethylamine, Water; 5 min, 25 °C Experimental Protocols	Convenient synthesis of 2-aminothiophene derivatives by acceleration of Gewald reaction under ultrasonic aqueous conditions By: Mojtahedi, Mohammad M.; et al Synthetic Communications (2010), 40(14), 2067-2074.
31-572-CAS-2169695 Steps: 1 Yield: 95% 1.1 Reagents: Triethylamine, Sulfur Solvents: Water; 7 h, rt Experimental Protocols	Efficient three-component Gewald reactions under Et₃N/H₂O conditions By: Abaee, M. Saeed; et al Journal of Sulfur Chemistry (2014), 35(3), 261-269.
31-572-CAS-6786282 Steps: 1 Yield: 95% 1.1 Reagents: Sulfur Catalysts: Calcium oxide Solvents: Ethanol; 1 h, reflux	A CaO catalyzed facile one-pot synthesis of 2-aminothiophenes using Gewald reaction By: Durgareddy, G. A. N. K.; et al Pharma Chemica (2013), 5(6), 294-298, 5 pp..

Scheme 17 (1 Reaction)

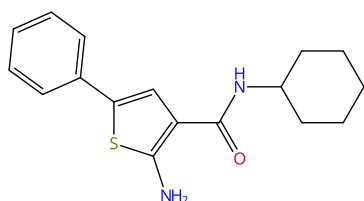
Steps: 1 Yield: 97%



Suppliers (85)

Suppliers (78)

Suppliers (69)



31-368-CAS-10712033 Steps: 1 Yield: 97% 1.1 15 min, rt 1.2 Reagents: S₈ Catalysts: Triethylamine Solvents: Water; 1 h, rt	A facile four-component Gewald reaction under organocatalyzed aqueous conditions By: Abaee, M. Saeed; et al ARKIVOC (Gainesville, FL, United States) (2014), (3), 1-10, 10 pp..
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Scheme 18 (1 Reaction)

Steps: 1 Yield: 97%



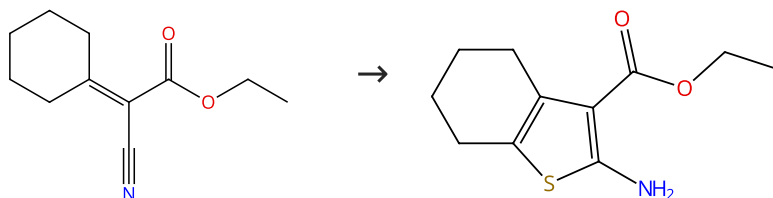
Suppliers (36)

Suppliers (40)

31-025-CAS-18122134	Steps: 1 Yield: 97%	Activation of elemental sulfur by electrogenerated cyanomethyl anion: synthesis of substituted 2-aminothiophenes by the Gewald reaction
1.1 Reagents: S ₈ Catalysts: Acetonitrile, Tetraethylammonium hexafluoro phosphate Solvents: Dimethylformamide; 1 min, rt; 1 h, rt		By: Feroci, Marta; et al
Experimental Protocols		Advanced Synthesis & Catalysis (2008), 350(17), 2740-2746.

Scheme 19 (2 Reactions)

Steps: 1 Yield: 96-97%



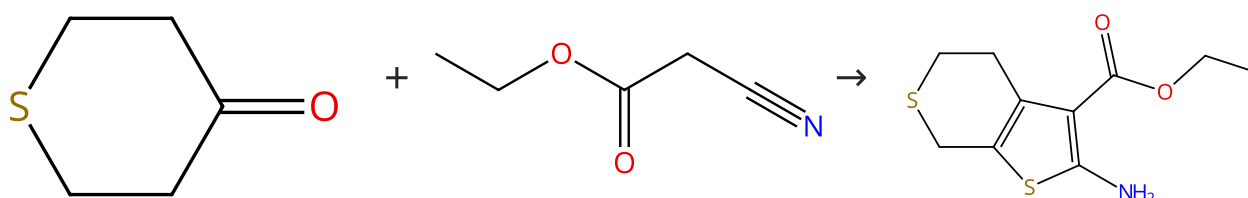
Suppliers (30)

Suppliers (77)

31-025-CAS-18142609	Steps: 1 Yield: 97%	Mechanistic observations in the Gewald syntheses of 2-aminothiophenes
1.1 Reagents: Diethylamine, Sulfur Solvents: Ethanol		By: Peet, Norton P.; et al
		Journal of Heterocyclic Chemistry (1986), 23(1), 129-34.
31-025-CAS-18142617	Steps: 1 Yield: 96%	Mechanistic observations in the Gewald syntheses of 2-aminothiophenes
1.1 Reagents: Morpholine, Sulfur Solvents: Ethanol		By: Peet, Norton P.; et al
		Journal of Heterocyclic Chemistry (1986), 23(1), 129-34.

Scheme 20 (2 Reactions)

Steps: 1 Yield: 95-97%



Suppliers (97)

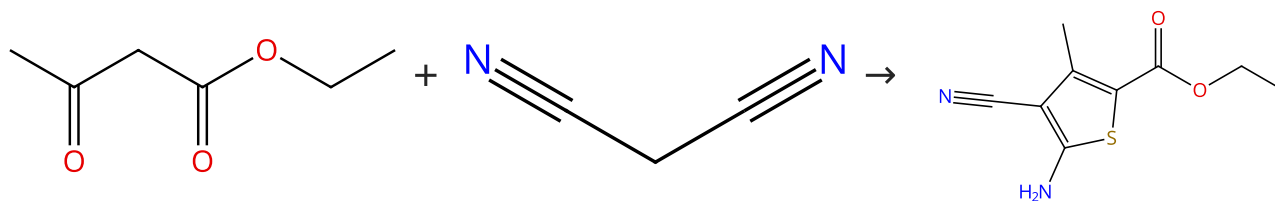
Suppliers (70)

Suppliers (41)

31-572-CAS-10540872	Steps: 1 Yield: 97%	Efficient three-component Gewald reactions under Et ₃ N/H ₂ O conditions
1.1 Reagents: Triethylamine, Sulfur Solvents: Water; 4 h, rt		By: Abaee, M. Saeed; et al
Experimental Protocols		Journal of Sulfur Chemistry (2014), 35(3), 261-269.
31-572-CAS-975489	Steps: 1 Yield: 95%	Aqueous DABCO, an efficient medium for rapid organocatalyzed Knoevenagel condensation and the Gewald reaction
1.1 Catalysts: Dabco Solvents: Water; 3 min, rt		By: Abaee, Mohammad Saeed; et al
1.2 Reagents: S ₈ ; 4 h, rt		Turkish Journal of Chemistry (2014), 38(4), 650-660.

Scheme 21 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (95)

Suppliers (63)

Suppliers (55)

31-572-CAS-19187785

Steps: 1 Yield: 97%

Green and efficient technique for the three component synthesis of 2-aminothiophenes using ball-milling method through Gewald reaction

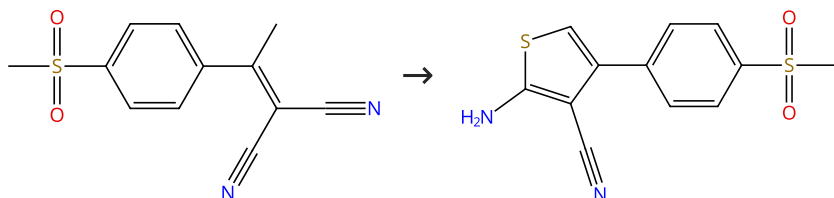
By: M'hamed, Mohamed Ould; et al
 Edited by Seijas, Julio A.; et al

International Electronic Conference on Synthetic Organic Chemistry, 20th, Nov. 1-30, 2016 (2016), 1-8.

1.1 Reagents: S₈; 30 min

Scheme 22 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (3)

31-025-CAS-18217736

Steps: 1 Yield: 96%

New conditions for the synthesis of thiophenes via the Knoevenagel/Gewald reaction sequence. Application to the synthesis of a multitargeted kinase inhibitor

By: Barnes, David M.; et al

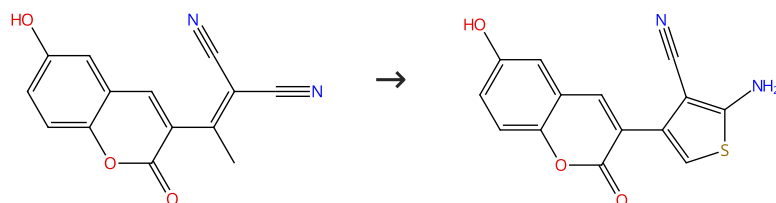
Tetrahedron (2006), 62(49), 11311-11319.

1.1 Reagents: Sulfur
 Solvents: Tetrahydrofuran; rt → 35 °C

1.2 Reagents: Sodium bicarbonate
 Solvents: Water; 1 h, 35 °C; 30 min, 35 °C

Scheme 23 (1 Reaction)

Steps: 1 Yield: 96%



Supplier (1)

Supplier (1)

31-025-CAS-22088503

Steps: 1 Yield: 96%

Improved one-pot synthetic conditions for synthesis of functionalized fluorescent coumarin-thiophene hybrids: Syntheses, DFT studies, photophysical and thermal properties

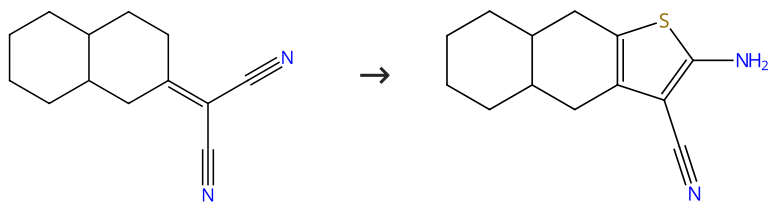
By: Yahaya, Issah; et al

Tetrahedron (2019), 75(14), 2143-2154.

1.1 Reagents: S₈
 Catalysts: Triethylamine
 Solvents: Ethanol; 3 min, 80 °C

Scheme 24 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (3)

Supplier (1)

31-025-CAS-18124143

Steps: 1 Yield: 96%

Activation of elemental sulfur by electrogenerated cyanomethyl anion: synthesis of substituted 2-aminothiophenes by the Gewald reaction

By: Feroci, Marta; et al

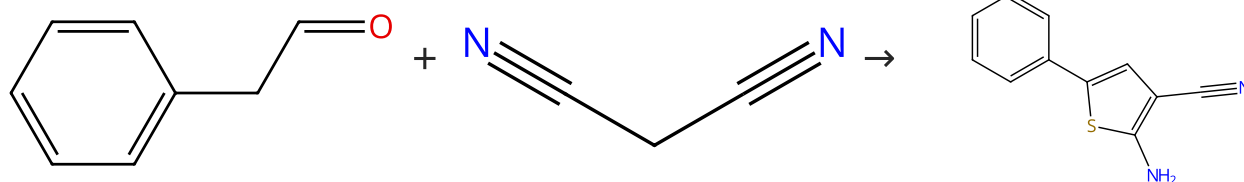
Advanced Synthesis & Catalysis (2008), 350(17), 2740-2746.

1.1 **Reagents:** S₈
Catalysts: Acetonitrile, Tetraethylammonium hexafluorophosphate
Solvents: Dimethylformamide; 1 min, rt; 1 h, rt

Experimental Protocols

Scheme 25 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (69)

Suppliers (63)

Suppliers (41)

31-572-CAS-3023344

Steps: 1 Yield: 96%

Microwave-assisted synthesis of 5-substituted 2-aminothiophenes starting from arylacetaldehydes

By: Revelant, Germain; et al

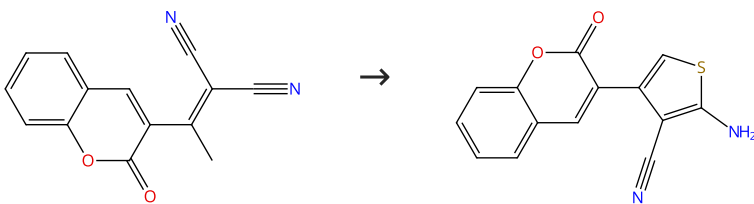
Synthesis (2011), (18), 2935-2940.

1.1 **Reagents:** Morpholine, S₈
Solvents: Ethanol; 20 min, 70 °C

Experimental Protocols

Scheme 26 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (7)

Suppliers (2)

31-025-CAS-21078640

Steps: 1 Yield: 96%

Improved one-pot synthetic conditions for synthesis of functionalized fluorescent coumarin-thiophene hybrids: Syntheses, DFT studies, photophysical and thermal properties

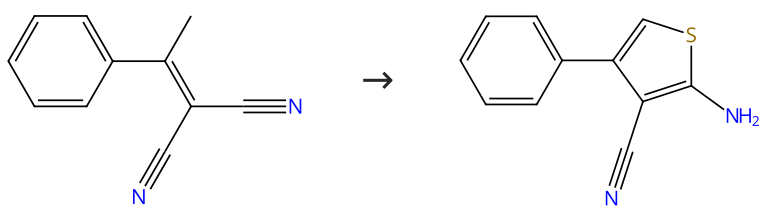
By: Yahaya, Issah; et al

Tetrahedron (2019), 75(14), 2143-2154.

1.1 **Reagents:** S₈
Catalysts: Triethylamine
Solvents: Ethanol; 3 min, 80 °C

Scheme 27 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (50)

Suppliers (42)

31-025-CAS-18213706

Steps: 1 Yield: 96%

New conditions for the synthesis of thiophenes via the Knoevenagel/Gewald reaction sequence. Application to the synthesis of a multitargeted kinase inhibitor

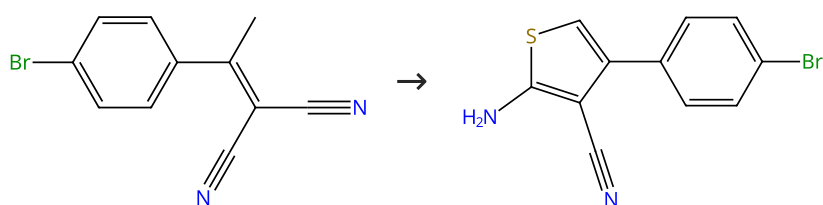
- 1.1 Reagents: Sulfur
Solvents: Tetrahydrofuran; rt → 35 °C
- 1.2 Reagents: Sodium bicarbonate
Solvents: Water; 1 h, 35 °C; 30 min, 35 °C

By: Barnes, David M.; et al

Tetrahedron (2006), 62(49), 11311-11319.

Scheme 28 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (7)

Suppliers (5)

31-025-CAS-18217358

Steps: 1 Yield: 96%

New conditions for the synthesis of thiophenes via the Knoevenagel/Gewald reaction sequence. Application to the synthesis of a multitargeted kinase inhibitor

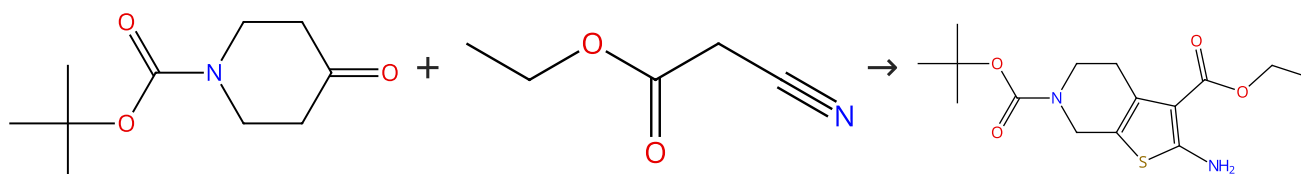
- 1.1 Reagents: Sulfur
Solvents: Tetrahydrofuran; rt → 35 °C
- 1.2 Reagents: Sodium bicarbonate
Solvents: Water; 1 h, 35 °C; 30 min, 35 °C

By: Barnes, David M.; et al

Tetrahedron (2006), 62(49), 11311-11319.

Scheme 29 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (55)

Suppliers (70)

Suppliers (66)

31-572-CAS-7647564

Steps: 1 Yield: 96%

An efficient one-pot synthesis of substituted 2-aminothiophenes via three-component Gewald reaction catalyzed by L-proline

- 1.1 Reagents: S₈
Catalysts: L-Proline
Solvents: Dimethylformamide; 20 h, 60 °C

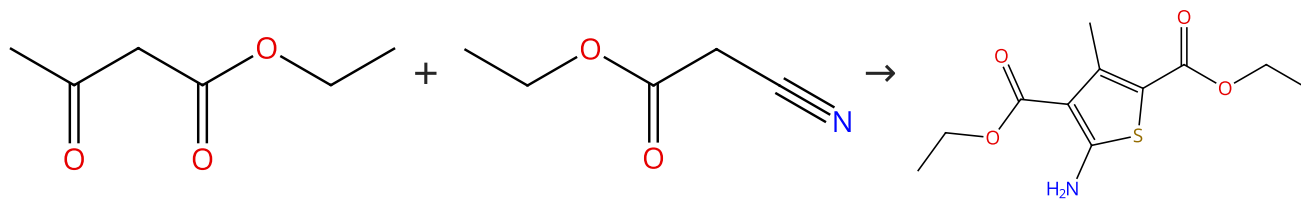
By: Wang, Tao; et al

Synlett (2010), (9), 1351-1354.

Experimental Protocols

Scheme 30 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (95)

Suppliers (70)

Suppliers (76)

31-572-CAS-19187787

Steps: 1 Yield: 96%

Green and efficient technique for the three component synthesis of 2-aminothiophenes using ball-milling method through Gewald reaction

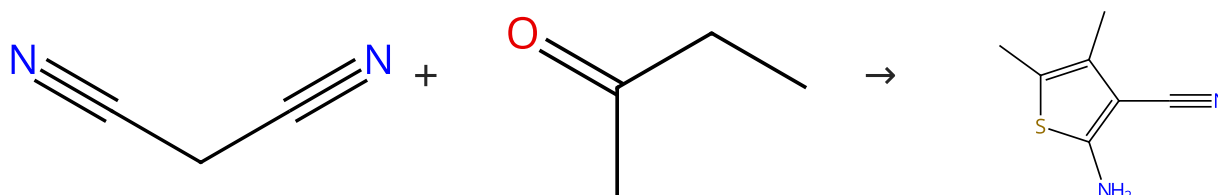
By: M'hamed, Mohamed Ould; et al
 Edited by Seijas, Julio A.; et al

International Electronic Conference on Synthetic Organic Chemistry, 20th, Nov. 1-30, 2016 (2016), 1-8.

1.1 Reagents: S₈; 30 min

Scheme 31 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (63)

Suppliers (116)

Suppliers (66)

31-572-CAS-6369637

Steps: 1 Yield: 96%

Bovine serum albumin-catalyzed one-pot synthesis of 2-aminothiophenes via Gewald reaction

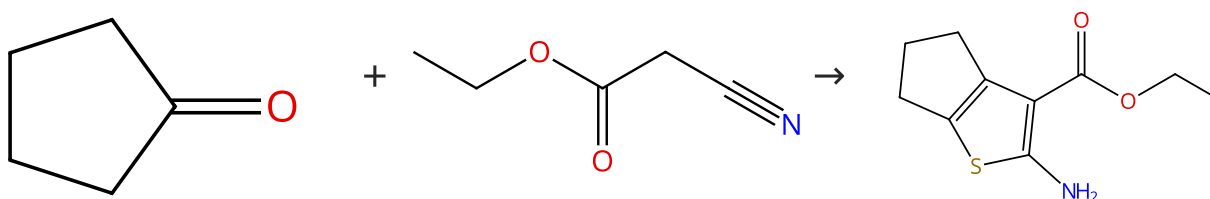
By: Zhao, Dan-Dan; et al

Journal of Molecular Catalysis B: Enzymatic (2013), 95, 29-35.

1.1 Reagents: Sulfur
 Solvents: Dimethylformamide; 4 h, 50 °C

Scheme 32 (2 Reactions)

Steps: 1 Yield: 95-96%



Suppliers (78)

Suppliers (70)

Suppliers (75)

31-572-CAS-4798522

Steps: 1 Yield: 96%

Convenient synthesis of 2-aminothiophene derivatives by acceleration of Gewald reaction under ultrasonic aqueous conditions

By: Mojtahedi, Mohammad M.; et al

Synthetic Communications (2010), 40(14), 2067-2074.

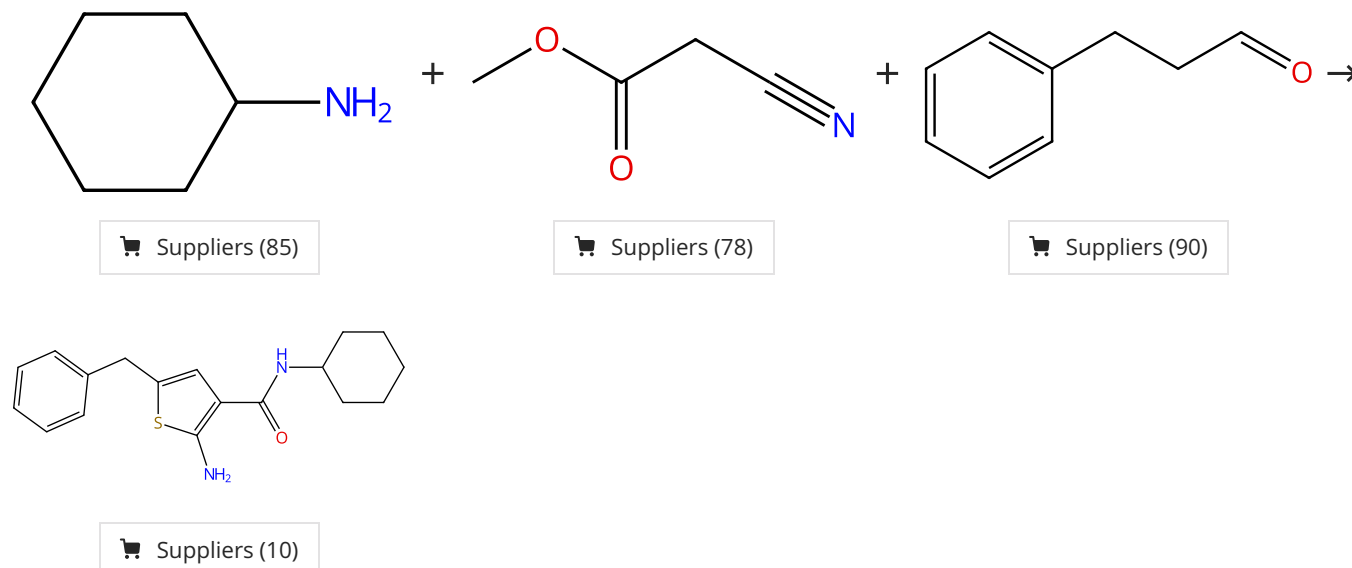
1.1 Reagents: S₈
 Solvents: Diethylamine, Water; 5 min, 25 °C

Experimental Protocols

31-572-CAS-4521650	Steps: 1 Yield: 95%	Efficient three-component Gewald reactions under Et ₃ N/H ₂ O conditions
1.1 Reagents: Triethylamine, Sulfur Solvents: Water; 7 h, rt		By: Abaee, M. Saeed; et al
Experimental Protocols		Journal of Sulfur Chemistry (2014), 35(3), 261-269.

Scheme 33 (1 Reaction)

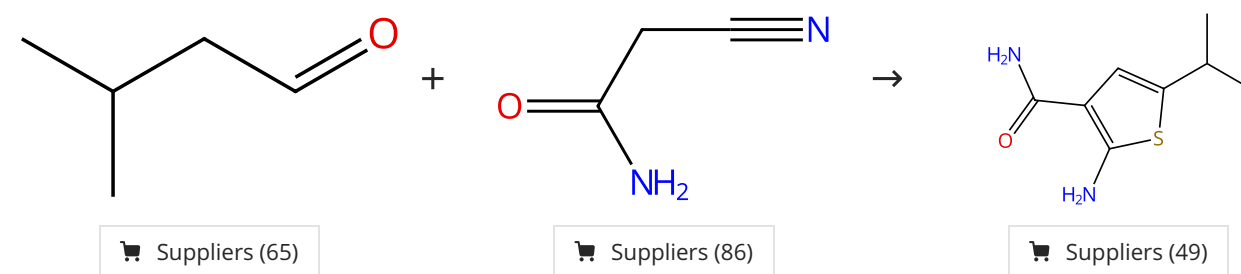
Steps: 1 Yield: 95%



31-368-CAS-11301256	Steps: 1 Yield: 95%	A facile four-component Gewald reaction under organocatalyzed aqueous conditions
1.1 15 min, rt 1.2 Reagents: S ₈ Catalysts: Triethylamine Solvents: Water; 45 min, rt		By: Abaee, M. Saeed; et al
		ARKIVOC (Gainesville, FL, United States) (2014), (3), 1-10, 10 pp..

Scheme 34 (1 Reaction)

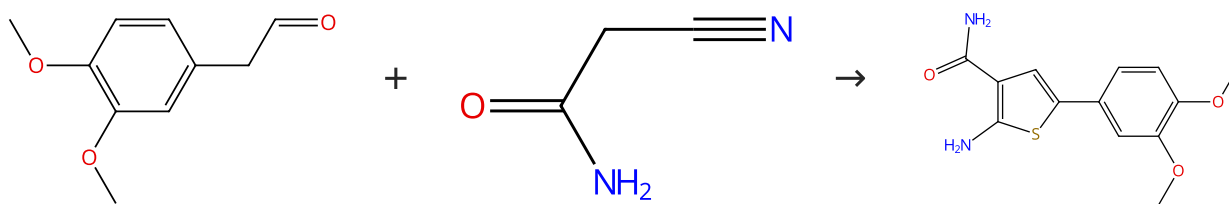
Steps: 1 Yield: 95%



31-614-CAS-24926732	Steps: 1 Yield: 95%	Microwave-assisted synthesis of 2-aminothiophene derivatives via improved gewald reactions
1.1 Reagents: Pyrrolidine, S ₈ Solvents: Dimethylformamide; 30 min, 50 °C		By: Ruan, Bankang; et al
Experimental Protocols		Heterocycles (2021), 102(10), 2007-2018.

Scheme 35 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (54)

Suppliers (86)

31-572-CAS-281443

Steps: 1 Yield: 95%

Microwave-assisted synthesis of 5-substituted 2-aminothiophenes starting from arylacetaldehydes

1.1 Reagents: Morpholine, S₈
Solvents: Ethanol; 20 min, 70 °C

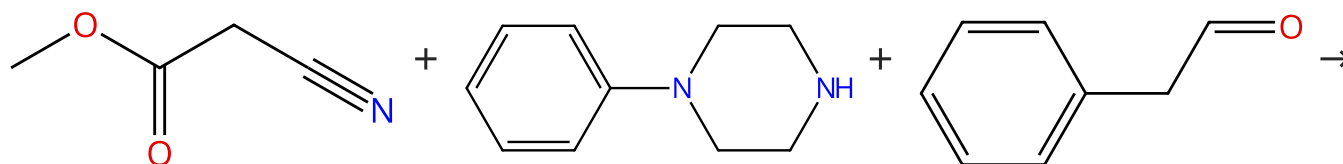
By: Revelant, Germain; et al

Experimental Protocols

Synthesis (2011), (18), 2935-2940.

Scheme 36 (1 Reaction)

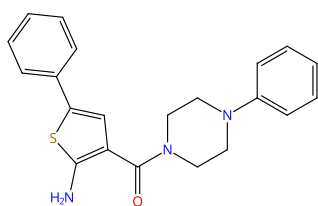
Steps: 1 Yield: 95%



Suppliers (78)

Suppliers (79)

Suppliers (69)



Supplier (1)

31-368-CAS-6737774

Steps: 1 Yield: 95%

A facile four-component Gewald reaction under organocatalyzed aqueous conditions

1.1 15 min, rt

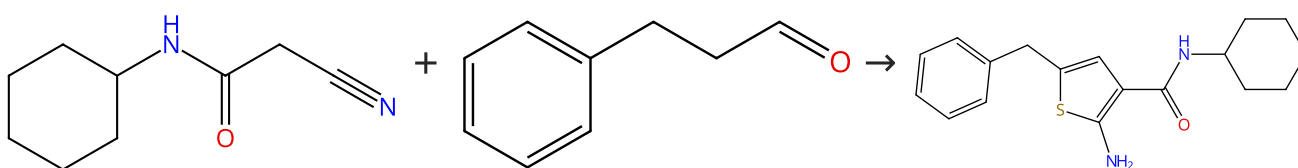
By: Abaee, M. Saeed; et al

1.2 Reagents: S₈
Catalysts: Triethylamine
Solvents: Water; 2 h, rt

ARKIVOC (Gainesville, FL, United States) (2014), (3), 1-10, 10 pp..

Scheme 37 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (50)

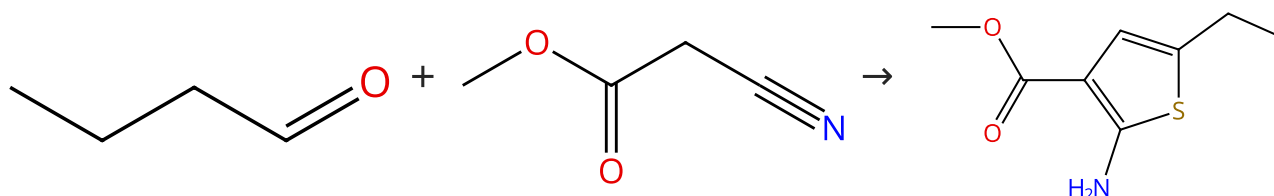
Suppliers (90)

Suppliers (10)

31-572-CAS-14745789	Steps: 1 Yield: 95%	Cyanoacetamide MCR (III): Three-Component Gewald Reactions Revisited
1.1 Reagents: Triethylamine, Sulfur Solvents: Ethanol; 10 h, 50 °C		By: Wang, Kan; et al
Experimental Protocols		Journal of Combinatorial Chemistry (2010), 12(1), 111-118.

Scheme 38 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (71)

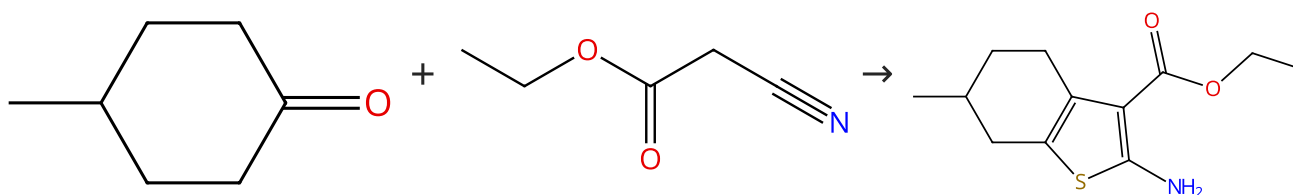
Suppliers (78)

Suppliers (61)

31-614-CAS-24926738	Steps: 1 Yield: 95%	Microwave-assisted synthesis of 2-aminothiophene derivatives via improved gewald reactions
1.1 Reagents: Pyrrolidine, S ₈ Solvents: Dimethylformamide; 30 min, 50 °C		By: Ruan, Bankang; et al
Experimental Protocols		Heterocycles (2021), 102(10), 2007-2018.

Scheme 39 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (77)

Suppliers (70)

Suppliers (65)

31-614-CAS-43798557	Steps: 1 Yield: 95%	Microwave-Assisted Synthesis of 2-Aryl-5,6,7,8-tetrahydrobenzo[4,5]thieno[2,3-d]pyrimidinones
1.1 Reagents: Triethylamine, S ₈ Solvents: Ethanol; 60 min, rt; overnight, reflux		By: Mehrabi, Hossein; et al
Experimental Protocols		ChemistrySelect (2024), 9(47), e202404000.